

Evolutionary Search Comparison with Selected-Frontier Audit Packages

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Abstract

This paper documents the BISECT evolutionary and Pareto comparison path. It separates frontier quality from selected-plan validity and requires fixed-seed metadata, validity-preserving operators, and RPLAN/RCTX audit packages for any selected frontier entry. A non-dominated entry is treated as an observed search result under a declared budget, not as a legal or policy preference by itself.

1 Introduction

U.19 documents `bisect-pareto` and `bisect-cli::pareto_cmd`, exposed through `bisect_pareto` and selected-frontier audit flags. The paper is an evolutionary/multi-objective implementation paper, not a proof that a computed frontier is globally complete.

The method belongs beside U.14 and U.15 rather than replacing them. U.19 can show that a frontier was generated under a seed, budget, and objective vector; U.14 supplies the selection record explaining why one non-dominated entry was chosen; U.15 states the legal or policy posture that may be attached to that choice. Once a frontier entry is selected, U.20's fixed-point audit path takes over: the selected plan must be exported as RPLAN/RCTX with certificate and manifest sidecars.

2 Implementation Contract

The implementation contract has three parts: validity-preserving crossover and mutation operators, deterministic fixed-seed frontier metadata, and selected frontier export. Operators may return valid children, explicit valid fallbacks, or rejected-child records; invalid silent outputs falsify the contract.

The atlas view makes the frontier decision explicit. Offspring creation, repair/fallback, frontier observation, and selected-index export are different events. A selected entry need not be best in every objective; it is a chosen trade-off point whose generation, objective vector, validity status, and selection index must travel with the RPLAN/RCTX package.

The implementation's frontier object makes the record concrete. Each frontier entry carries a one-based district assignment, objective values for edge cut, seats, and VRA

deficit, a non-dominated flag, generation found, plan-level validity status, and optional audit-certificate identity fields for exported entries. The final NDJSON line records population size, generation count, frontier size, base seed, runtime, output version, and a SHA-256 hash of the plan lines. This is the evidence that a row was observed in a particular search run, not evidence that no better row exists outside that budget.

A selected-frontier example is deliberately small. Suppose index 1 has edge cut 137 and county splits 8 while index 0 has edge cut 130 and county splits 13. Index 1 is not globally “better”; it is the selected trade-off. Exporting it therefore records selected index 1, frontier size, base seed, generation, objective vector, and validity status before writing the selected RPLAN/RCTX package.

Layer	Record	Claim boundary
Operator-valid	parent ids, seed, parameters, validity status	child validity
Operator-fallback	parent ids, failed operator, fallback reason	validity preservation
Frontier-recorded	objective vectors and dominance metadata	search result under budget
Selected-packaged	chosen entry and audit package id	auditable final plan
Selection-deferred	frontier entry without selection record	comparison only

Table 1: Evolutionary comparison record layers.

Artifact field	Current value family	Interpretation
Frontier row	plan, objectives, generation, validity	Observed comparison entry
Metadata row	population, generations, base seed, hash	Reproducible run envelope
Selected package	selected index, frontier size, RPLAN/RCTX paths	Chosen entry exported for audit
Lineage extra	method, seed, generation, objectives, validity	Bridge from package back to frontier
Audit certificate	profile, context hash, result, content hash	Final-plan validity evidence

Table 2: Concrete U.19 fields that separate frontier evidence from audit evidence.

The implementation therefore separates producing candidates, observing a frontier, and selecting an entry for downstream audit. Frontier membership is a search-output claim. Selected-package status is an audit-readiness claim. Selection-deferred status means the entry can be compared but should not be reported as a chosen plan without the U.14 selection record.

3 Audit Artifacts

A selected frontier entry should be exportable as RPLAN/RCTX with certificate and manifest sidecars. The audit package verifies the selected plan and lineage under a declared profile; it does not certify that the frontier is complete or that the selected trade-off is legally preferred.

The manifest must include the fixed seed, population and generation identifiers, operator parameters, parent ids for generated descendants, selected entry id, objective vector, constraints profile, RPLAN path, RCTX path, certificate path, and verification transcript. If a candidate reached the frontier through a fallback or repair path, that path remains part of the selected-entry lineage.

Selection criteria are separate audit inputs. U.19 can package a selected frontier entry for verification, but preference among valid non-dominated entries belongs to the U.14 selection record and the U.15 posture boundary.

The public golden package can be checked either through the RPLAN certificate verifier or through the BISECT manifest verifier.

The selected package schema is `bisect-pareto-selected-frontier-package-v1`. It writes `selected-frontier.rplan`, `selected-frontier.rctx`, `audit-certificate.json`, and `manifest.json`. The manifest records seed, selected index, frontier size, objective values, validity status, certificate digest, certificate content hash, legal profile id, and context hash. That package is the point where an exploratory evolutionary row becomes a normal audit object.

4 Evaluation Plan

Current evidence includes L0/L1 crossover, mutation, determinism, selected package verification tests, and a public U.19 selected-frontier package in the package corpus. Seed/objective sensitivity, VRA-aware objective disclosure, and frontier-quality comparisons remain future empirical work.

Claim	Current evidence	Publication gap
Operators preserve validity or emit fallbacks	L0 crossover/mutation tests	Operator transcript examples
Fixed-seed frontier records are reproducible	L0/L1 determinism tests	Seed sensitivity appendix
Selected entries can be audited as RPLAN/RCTX packages	Selected package verification tests plus public manifest and transcript	larger frontier export
Frontier quality is robust across objectives and budgets	Future evidence only	Benchmark and sensitivity protocol
Selected trade-off is legally or policy preferred	Out of scope for U.19	U.14 selection record and U.15 posture analysis

Table 3: Claim and evidence ladder for U.19.

Command or package	Evidence produced
<code>cargo test -p bisect-pareto</code>	Checks deterministic frontiers, non-domination, operator validity/fallback behavior, NDJSON metadata, and selected-package sidecars.
<code>bisect pareto --selected-frontier-index ...</code>	Writes the frontier NDJSON and selected RPLAN/RCTX/audit/manifest package for the chosen row.
Public U.19 golden package	Preserves a selected-frontier package with manifest and verification-ready sidecars.

Table 4: Current U.19 evidence path.

The representative CLI shape is:

```
bisect pareto --state TT --year 2020 \
--selected-frontier-index 1 \
--selected-frontier-context context.rctx \
--selected-frontier-out selected-frontier
```

Changing the base seed, objectives, population size, or generation budget can change the frontier and therefore the meaning of index 1. Those parameters are part of the evidence, not incidental run decoration.

5 Limitations

Search budget affects frontier quality. A valid selected plan does not imply a complete frontier, and a non-dominated entry does not settle policy or legal choice without the U.14 selection record and U.15 posture boundary. Fixed-seed reproducibility is necessary for audit but does not by itself imply fairness, robustness, or legal adequacy.

Future empirical work must report budgets, seeds, objective weights or definitions, operator parameters, fallback rates, and sensitivity to VRA-aware inputs before making broader quality claims. The public selected-frontier package proves that a chosen entry can be audited; it does not prove that the selected entry is robust to seed choice, objective definition, or search budget. A publication-quality comparison should include a seed/objective sensitivity appendix before using frontier shape as substantive evidence.

6 Conclusion

U.19 makes evolutionary comparison compatible with the fixed point: exploratory frontiers remain heuristic search objects, while selected outputs become auditable plan packages. The paper treats operator validity, frontier observation, selection, and audit packaging as separate layers, preserving room for policy choice without hiding the computational lineage of selected plans.

References

BISECT project. U.19 evolutionary search comparison paper plan, 2026. Local planning document and implementation roadmap.